Dylan Thornburg

Instructions

- This assignment assesses your understanding of logistic regression and its application in real-world safety analysis.
- Download the paper: applsci-09-03449, to answer questions on applications of logistic regression.
- Submit a single PDF file.
- Total points: 30

Part 1: Understanding the Model (10 points)

1. (2 pts) What is the dependent variable in this logistic regression study? Why is logistic regression appropriate?

The dependent variable is post-incident state and log reg is appropriate b/c the variable is binary.

2. (4 pts) What does an odds ratio mean in this study? Explain, using the example where the odds ratio for fall/slip/trip injury is 1.86.

An odds ratio for this study is the likelihood someone gets disabled from an injury compared to the reference data. An odds ratio of 1.86 means a worker is 1.86 times more likely to be in a disability post-incident state from a fall/slip/trip.

3. (2 pts) Define *post-incident state*. How was it categorized in the dataset?

Post-incident refers to the state of the worker after the incident.

Two categories; medical state (no time off)
disability state (disability time off)

4. (2 pts) Identify two significant predictors from the study. Briefly describe their effect on the probability of a disability outcome.

- Type of injury.
- Occupation

Based on these predictors, the odds of a disability state can increase or decrease drastically.

Part 2: Applying the Model (10 points)

5. (4 pts) A 60-year-old worker with 10 years of experience suffers a lower extremity injury due to a fall while working in a grain elevator. Using the coefficients from Table 5 and the logistic regression formula in the paper, calculate the predicted probability of a disability outcome.

int age exp grain (ind) elevator lower extremity fall grain elevator (occ)

$z = -2.12 + .02(60) - .02(10) - .05 + .49 + .62 + .32$

$= .26$ $\quad e^{-z} = .771$ $\quad \frac{1}{1+.771} = \boxed{.565}$

6. (3 pts) Change the above scenario to a head injury caused by a puncture. How does the prediction change?

$z = -2.12 + (.02)60 - .02(10) - .05 + .32 - .75 - 1.15$

$z = -2.75$ $\quad e^{-(-2.75)} = 15.643$ $\quad \frac{1}{1+15.643} = .06$

The probability of the disability outcome is much lower now.

7. (3 pts) What are some limitations of the logistic model in this study? Suggest a possible improvement.

We don't have good health indicators for the workers (besides maybe age). Implement sensors to see who is more susceptible to a disability state post injury.

Part 3: Critical Reflection (10 points)

8. (3 pts) How does logistic regression help identify high-risk occupations in the study? Why is this valuable?

It assigns odds ratios to occupations based on past injury data. This is valuable so we can know which jobs tend to be the most dangerous.

9. (3 pts) What is the Wald chi-squared test, and how was it used in this paper?

The Wald square test determines if a predictor has a meaningful effect on outcome (statistically diff from 0). In this study, all input variables were evaluated to make sure they were significant. All were ($p < .05$).

10. (4 pts) You are the safety manager in an agribusiness firm. Based on the study's findings, provide two actionable recommendations to reduce disability outcomes.

Falls, slips, trips, had a high B. I would try to reduce slippery surfaces while keeping the floors smooth (no bumps to trip people). I also would try to add infrastructure to help older employees (rails, protective gear, etc.).